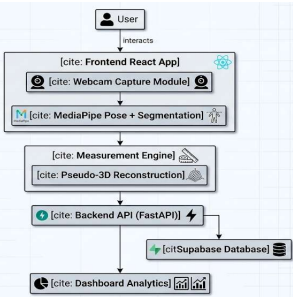
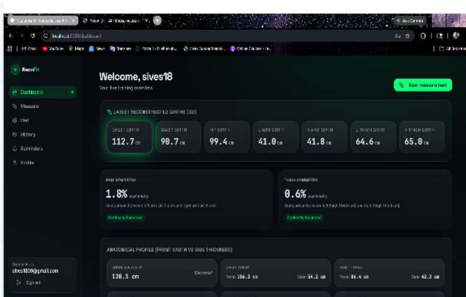
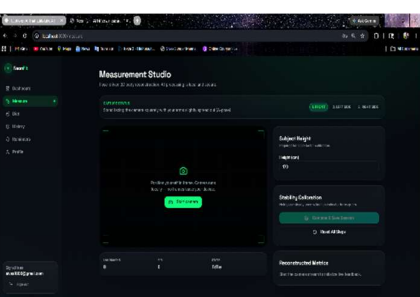


AI-Based Multi-View Muscle Hypertrophy Tracking and Fitness Analytics System Using Computer Vision		
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Keywords: Muscle Hypertrophy, Computer Vision, MediaPipe Pose, Body Measurement, Multi- view Geometry, Fitness Tracker, AI Based Analysis.		
Abstract. The primary goal of this project involves developing an AI-based muscle hypertrophy tracking system that automatically measures body dimensions and monitors muscle growth over time. Fitness enthusiasts and individuals engaged in strength training often face challenges in accurately tracking their physical progress, as traditional methods rely on manual measurements and subjective visual comparisons, which lack consistency and precision. With the rapid advancement of artificial intelligence and computer vision, there is an increasing demand for intelligent systems that can support objective and data-driven fitness analysis. The proposed system utilizes computer vision techniques to extract anatomical landmarks from real-time images and convert them into meaningful body measurements. The process begins with capturing multi-view images (front and side), followed by pose estimation using frameworks such as MediaPipe Pose to detect key body points. These landmarks are then processed to compute measurements such as arm length, shoulder width, and thigh length. A scaling mechanism is applied to convert pixel based values into real-world units, ensuring accurate and consistent results. A key feature of the system is the use of multi-view geometric modeling, where width and thickness information from different perspectives are combined to approximate muscle size. This is achieved through a pseudo-3D approach, where muscle cross-sections are modeled using an elliptical formulation. The system also incorporates validation mechanisms to ensure correct posture during capture, thereby improving measurement reliability. The experimental results demonstrate that the system produces consistent measurements with low error margins and effectively tracks muscle growth trends over time. The solution operates efficiently using standard hardware and provides repeatable results, making it suitable for practical fitness applications as well as academic purposes.		
<pre>graph TD User((User)) -- interacts --> Frontend[Frontend React App] Frontend --> Webcam[Webcam Capture Module] Webcam --> MediaPipe[MediaPipe Pose + Segmentation] MediaPipe --> Measurement[Measurement Engine] Measurement --> Pseudo3D[Pseudo-3D Reconstruction] Pseudo3D --> Backend[Backend API FastAPI] Backend --> Supabase[Supabase Database] Supabase --> Dashboard[Dashboard Analytics]</pre>  		

Conclusion: In summary, NeoFit demonstrates how affordable AI-assisted measurement, pseudo-3D reconstruction, and automated analytics can converge to enable longitudinal fitness tracking, marking a significant contribution at the intersection of computer vision and fitness analytics.